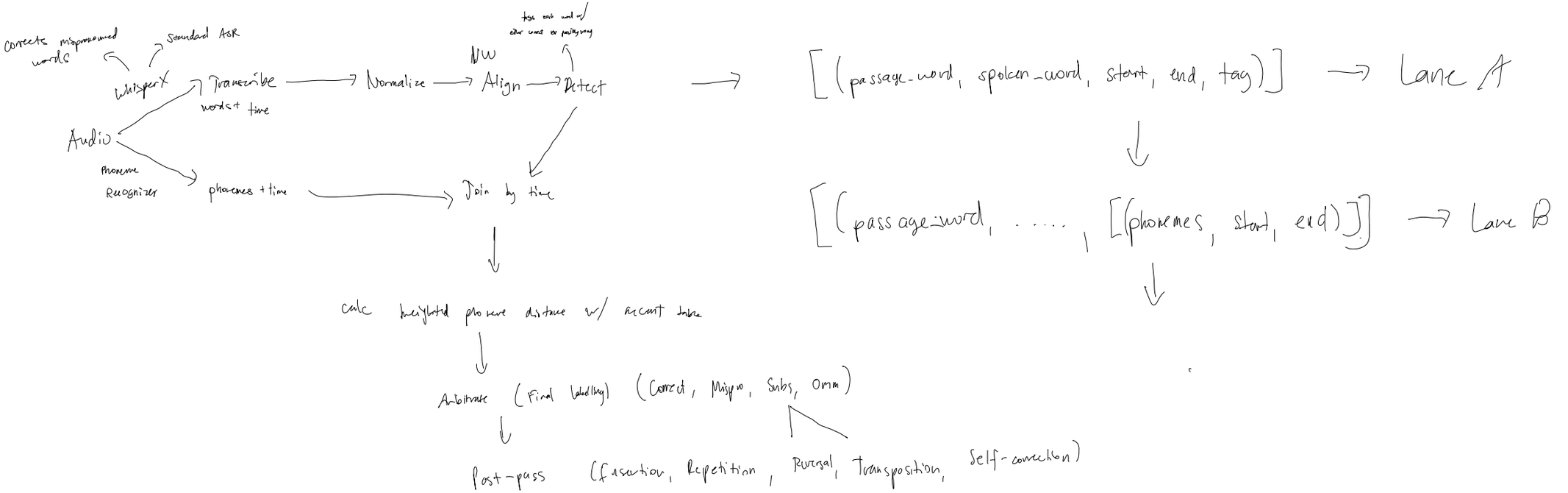
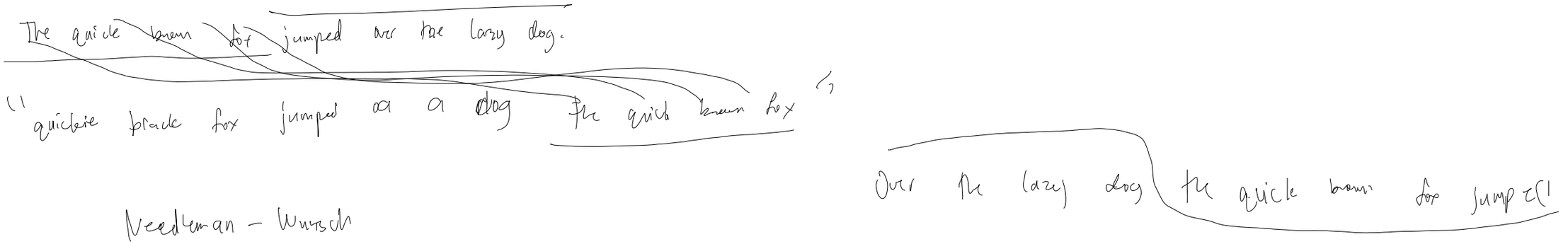
AI

Saturday, 31 October 2026 2:20 PM



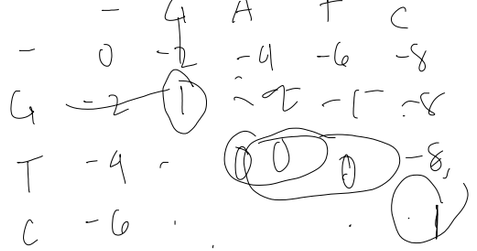


Needleman - Wunsch

GATC
GTC

res { +1 Match
-1 Mismatch
-2 Gap

-3



max(T2 + res, up + gap, left + gap)

G A T C
| | | |
G T C

Passage: Every Saturday, Manuel goes to market with his father, Mang Ador, happily.

① " [Um] Every Every Sat nuelMa [gus] goose to ∧ the store ∧ width with see Mang Ador father. " ⟵

Mispronunciation: goes → goose (gus phonetic spelling)

Omission: happily → x

Substitution: market → store

Insertion: x → the

Repetition: Every → Every Every

Transposition: father, Many /fɑðər/ → Many Ador, father

Reversal: his → see (sih phoenetic spelling)

Self-correction: with → widthwith

Passage: The small girl saw a bad dog near the slide. She was happy to pet the brown puppy at the park.
Actual Audio: The girl saw a dub cat near the slid. She happy was to pet to pet the very brown puppy at the pork park.

Omission: small -> x
Reversal: bad -> dub (phoenetic dab)
Substitution: dog -> cat
Mispronunciation: slide -> slid (phoenetic sleed)
Transposition: was happy -> happy was
Repetition: to pet -> to pet, to pet
Inserstion: x -> very
Self-correction: park -> pork park

1. Transcription: t = [(word, start, end)]
2. Normalization: the girl saw a dub cat near the slid she happy was to pet to pet the very brown puppy at the pork park; n = [(word, start, end)]
3. Alignment: a = (passage, n) -> [(passage_word, spoken_word, start, end)] {passage_word and spoken_word are nullable)]
4. Detection: a -> [(passage_word, spoken_word, start, end, pre_label)] {correct, omission, insertion, substitution,

FE dependencies

/login → none

/register → none

/dashboard → reading_score, actual_phone, spoken_phone, audio_link, actual_phoneme, spoken_phoneme, wpm, is_word_by_word, is_disregard_punctuation, is_monotonous, is_lack_expression, is_finger_pointing,

miscue_type ← read_start, read_end

[reading_level]

for a single